Universität Trier, Fachbereich II
Professur Computerlinguistik

On the Effects of Quantization of Qwen3 and Qwen3.5 in Consumer-Scale Deployment

Trends in Machine Learning(12401626)

Master's in Natural Language Processing

submitted by

Emrehan Dalaman (1873271)

s2emdala@uni-trier.de

Dhanush Kannan (1853946)

s2dhkann@uni-trier.de

Submission: March 31, 2026

Supervisor: Prof. Kai Kugler, M.A

UNIVERSITÄT TRIER

Declaration of Academic Integrity

I,

Last name, first name: Emrehan Dalaman, Dhanush Kannan

Student ID number: 1873271, 1853946

hereby declare that I have written this thesis independently and have not used any sources or aids other than those specified. All direct or indirect content and ideas from external sources are acknowledged as references. I confirm that I have not previously submitted this thesis in part or whole for any other academic credit nor has the work been submitted to any other examination office and has not been published.

I confirm that I have only used text or other content-generating aids based on generative artificial intelligence (GenAI) in the manner permitted in writing by the examiner and that I have explicitly marked GenAI-generated passages.
When using GenAI tools, I have indicated these with their product name and listed them in full in an appendix.

I am fully responsible for any GenAI-generated passages and content I have used in my work.

Thesis title: On the Effects of Quantization of Qwen3 and Qwen3.5 in Consumer-Scale Deployment

Programme: Master's in Natural Language Processing

Examiner: Prof. Kai Kugler, M.A

This is an achievement for a

- ☐ Bachelor's thesis
- ☐ Master's thesis
- ☑ other examination achievement for module (term paper etc.)

Trier, 31/03/2026
Place, Date

Emrehan Dalaman, Dhanush Kannan
Signature

Stand: 06/2025

Contents

1 Introduction **2**

2 Background and Related Work **3**

2.1 Positional Decay and Long-Context Failures 3

2.2 Hybrid Architectures 3

2.3 Post-Training Quantization 3

3 Methodology **4**

3.1 Models and Quantization 4

3.2 Benchmarks and Experimental Grid 4

3.3 Scoring and Refusal Detection 4

3.4 Hardware and Controls 5

4 Results **5**

4.1 Finding 1: Asymmetric Context-Length Sensitivity Under Quantization (RQ1) 5

4.2 Finding 2: The Q4 Failure Mode (RQ2) 6

4.3 Finding 3: Short-Context Controls Confirm Domain Specificity 7

4.4 Finding 4: Behavioral Divergence in Refusal Rates and Failure Modes (RQ3) 7

4.5 Finding 5: UD-Q3 Over-Performance as Evaluation Artifact and the Prompt-Ablation Result 8

5 Discussion **9**

5.1 The Confounding Variables 9

5.2 Practical Implications for Practitioners 10

6 Conclusion **10**

6.1 Limitations 11

6.2 Future work 11

References **12**

Appendices **13**

A Code Repository 13

B AI Usage Declaration 13

Abstract

Research in models that utilize efficient memory architectures such as NVIDIA's Nemotron 3 with Mamba-2 layers and Qwen3.5 with GDN layers shows that short context retrieval is almost a solved task. On the other hand, local deployment complicates the narrative, as recurrent memory accumulates errors when quantized, even though each architecture utilizes different strategies, like gating, to let the error decay. Even though a decaying memory means worse long-context degradation under aggressive quantization, SOTA models show promising results. This study tests that intuition empirically by evaluating Qwen3-4B quantizations, a multi-head Transformer model, and Qwen3.5-4B, a hybrid architecture replacing 75% of attention layers with Gated Delta Network (GDN) layers, across four GGUF quantization levels (Q8, Q4, Q3, UD-Q3-XL) on NeedleBench (Li et al., 2024) at 4k, 8k, and 16k token contexts.

On quantization error compounding, our results show the opposite of the expected direction: at Q4 precision on Multi-Needle Retrieval, Qwen3 loses 18.0 percentage points of accuracy from 4k to 16k context, while Qwen3.5 shows zero degradation under identical conditions. Secondary findings include a failure where 4-bit quantization underperforms 3-bit quantization for Qwen3, and a prompt-ablation study revealing that the Transformer-based model refuses to answer, and it is not a genuine capacity exhaustion.

Since the tested models differ not just in their architecture, but also in their training pipeline, RLHF, and alignment, one cannot make architectural claims; only under the conditions evaluated, Qwen3.5 shows better quantization resilience than Qwen3.

1 Introduction

The use of LLMs on consumer-grade hardware utilizes heavy quantization and KV cache compression. With advancements in the field like Unsloth's dynamic quantization schemes (Daniel Han and team, 2023) and Google's TurboQuant (Zandieh et al., 2025), it is easier than ever to run intelligent models locally with limited VRAM.

Recent hybrid architectures, including Gated Delta Networks (S. Yang et al., 2025) and Mamba (Gu and Dao, 2023), use fixed-memory recurrent hidden states, compressing history into hidden states rather than a growing key-value cache. A natural prediction follows: if quantization introduces errors into recurrent state updates, those errors may compound step-by-step, producing worse long-context degradation than in pure-attention Transformers, where each position's computation is independent (NVIDIA, 2025, §4.3). This study was motivated by not seeing that prediction in data. Our experimental findings turn the expected direction:

> At Q4 precision on Multi-Needle Retrieval, Qwen3 loses 18.0 percentage points of accuracy as context extends from 4k to 16k tokens, while Qwen3.5 shows zero degradation under identical conditions.

RQ1: How does weight quantization interact with context-decay profiles of Qwen3.5 versus Qwen3 across 4k-16k token contexts?

RQ2: How does quantization scheme, uniform versus dynamic, influence failure modes, and are these modes architecture-dependent?

RQ3: Do the two architectures exhibit qualitatively different behavioral failures (refusal rates, depth-position sensitivity) under increasing quantization pressure?

The study is restricted to the 4B parameter scale and 16k context length on a single consumer GPU for two reasons: small models make quantization errors more visible, and end users deploying locally require results that reflect their actual hardware constraints.

2 Background and Related Work

2.1 Positional Decay and Long-Context Failures

Studies on this case demonstrated that Transformer models retrieve facts placed at the beginning or end of a document more reliably than facts placed in the middle, coined as "lost in the middle" effect (Liu et al., 2024). Independent of positional bias, models also suffer from context overload: token volume degrades retrieval ability even without positional bottlenecks. The NeedleBench framework (Li et al., 2024) allows independently varying context length and needle insertion depth, making it well suited to this study's design.

2.2 Hybrid Architectures

Standard multi-head attention scales quadratically in compute, which led research into linear alternatives. Mamba (Gu and Dao, 2023) introduced selective state-space models, and recent architectures implemented the gated update rule (Dao and Gu, 2024). The Gated Delta Network (S. Yang et al., 2025) extended this line with the delta rule for memory updates, enabling the correction of specific memory associations rather than adding all key-value pairs indiscriminately. 75% of layers of Qwen3.5-4B are GDN layers operating at linear cost, while 25% are standard grouped-query attention layers (Qwen Team, 2026). Qwen3-4B uses grouped-query attention in all layers (A. Yang and the Qwen Team, 2025), providing a clean architectural baseline. Although full isolation of the architectural variable remains an open challenge, we expect Qwen3.5 to behave differently under quantization, because retrieval is exclusively handled by attention layers (Michalak and Abreu, 2025)

2.3 Post-Training Quantization

Recent work confirms that 4-bit quantization causes substantial long-context retrieval degradation in some configurations, while 8-bit methods largely preserve accuracy (Mekala et al., 2025). The GGUF ecosystem implements several schemes: Qn_K_M has uniform n-bit quantization across its layers; UD-Qn_K_XL (Unsloth Dynamic Daniel Han and team, 2023) preserving embeddings and sensitive layers at higher precision while compressing less sensitive layers to 3 bits. The comparison between uniform and dynamic schemes is central to this study.

3 Methodology

3.1 Models and Quantization

Two base models were evaluated: Qwen3-4B, and Qwen3.5-4B, across 4 quantizations for each model, Q8_0 (8-bit symmetric, near-full-precision reference baseline), Q4_K_M (uniform 4-bit M), Q3_K_M (uniform 3-bit M), and UD-Q3_K_XL (Unsloth Dynamic non-uniform 3-bit XL). This yields eight models to run experimental benchmarks on.

Model comparisons at Q3 use Q3_K_M for both models only. UD-Q3_K_XL is a distinct quantization method and is treated throughout as a separate configuration study. Tables mark UD-Q3 rows with † to prevent confusion.

3.2 Benchmarks and Experimental Grid

We tested 3 benchmarks with different roles. WikiText-103 perplexity (2,048-token sliding window) is the short-context sanity check: degradation with decreasing precision confirms all eight configurations operate correctly. GSM8K accuracy (250-problem sample, 2,048-token context) is the short-context reasoning control, showing that Qwen3 is not a globally inferior model as it excels at mathematical reasoning and that long-context failures are domain-specific.

NeedleBench Li et al., 2024 is our main benchmark. The model is prompted to retrieve a needle buried inside a document at a specified depth. Three tasks are evaluated: Single-Needle Retrieval (S-RT), Multi-Needle Retrieval (M-RT, the primary comparison), and Multi-Needle Reasoning (M-RS). Each is tested at 4k, 8k, and 16k tokens, with needles at six depth positions (5%, 10%, 30%, 50%, 70%, 90%) and 15 samples per depth-context combination, yielding 6,480 total evaluations. M-RT is the primary comparison because both models achieve nearly identical results at 4k (∼98-100%), minimizing baseline capability bias.

3.3 Scoring and Refusal Detection

Response correctness is a score that takes the maximum of Levenshtein similarity, coverage ratio, and exact string match, with a response considered correct if this maximum is greater than or equal to 0.5. Refusals are detected by checking for the

string "NOT FOUND" in the answer tags as the evaluation prompt instructs models to say when the target needle cannot be located. The refusal rate is the proportion of responses containing this string, and serves as a pointer for instruction-following degradation under quantization pressure.

3.4 Hardware and Controls

All evaluations ran on an NVIDIA RTX 4060 8 GB GPU. To prevent out-of-memory failures at 16k context, Flash Attention was enabled, execution ran in headless to be able to fit the context without out of memory errors, (recovering ∼900 MiB of VRAM), and chunked prefill was set to $n_{\text{batch}} = 256$. Thinking mode was disabled to compare retrieval behavior rather than chain-of-thought generation. Random seeds were fixed per depth condition, holding haystack content constant while varying only needle insertion position.

4 Results

Findings 1 and 2 inspect RQ1 (architecture × quantization interaction); Finding 3 isolates failures to specific domains; Finding 4 addresses RQ3 (behavioral differences); Finding 5 identifies an evaluation artifact.

4.1 Finding 1: Asymmetric Context-Length Sensitivity Under Quantization (RQ1)

The central finding is an asymmetry in how the two models degrade as context length increases under identical quantization. Table 1 presents Multi-Needle Retrieval (M-RT) accuracy, the cleanest comparison because both models achieve near-ceiling accuracy at 4k, isolating degradation as our variable. All quantizations use K_M variants; Qwen3/3.5 UD-Q3-XL employs a distinct per-layer dynamic allocation scheme (Daniel Han and team, 2023) and is treated as a separate condition.

Table 1: M-RT accuracy (%) by context length and quantization level. Qwen3 degrades sharply from 4k to 16k context after Q4 quantization, while Qwen3.5 remains stable.

Model / Quant.	**4k**	**8k**	**16k**	**4k→16k Δ**
Qwen3 Q8	99.8%	97.6%	96.2%	−3.6 pp
Qwen3 Q4	94.9%	87.1%	76.9%	−18.0 pp **!**
Qwen3 Q3	98.0%	95.6%	81.8%	−16.2 pp
Qwen3 UD-Q3†	100.0%	99.3%	92.9%	−7.1 pp
Qwen3.5 Q8	98.7%	98.4%	98.7%	0.0 pp
Qwen3.5 Q4	98.7%	98.7%	98.7%	0.0 pp ✓
Qwen3.5 Q3	97.8%	90.7%	92.7%	−5.1 pp
Qwen3.5 UD-Q3†	97.8%	98.4%	98.2%	+0.4 pp

At Q4 precision, Qwen3 falls 18.0 pp from 4k to 16k; Qwen3.5 is flat at 0.0 pp under identical conditions. At Q3, the same pattern is visible: Qwen3 −16.2 pp; Qwen3.5 −5.1 pp. Under Q8, both are robust (Qwen3 −3.6 pp). The same asymmetry appears on Multi-Needle Reasoning Task (Table 2): Qwen3 Q3 drops 22.2 pp (63.3% → 41.1%), while Qwen3.5 Q3 is still close to baseline. The M-RS comparison is complicated by a large Q8 baseline gap (Qwen3: 48.9%; Qwen3.5: 80.0%), which likely reflects training differences.

Table 2: M-RS accuracy by context length. The large Q8 baseline gap (48.9% vs. 80.0%) reflects training differences

Model / Quant.	**4k**	**8k**	**16k**	**4k→16k Δ**
Qwen3 Q8	48.9%	51.1%	46.7%	−2.2 pp
Qwen3 Q3	63.3%	57.8%	41.1%	−22.2 pp
Qwen3.5 Q8	80.0%	76.7%	75.6%	−4.4 pp
Qwen3.5 Q3	72.2%	73.3%	75.6%	+3.4 pp

4.2 Finding 2: The Q4 Failure Mode (RQ2)

An observation in the Qwen3 data: Q4 quantization produces worse retrieval performance than Q3 across multiple tasks and context lengths (Table 3). This does not apply to Qwen3.5, which degrades smoothly across precision levels. We hypothesize that the tested Q4 quantization has artifacts on Qwen3's instruction following abilities, causing the model to answer "NOT FOUND", where an ablation experiment reveals that the model does this even though it knows the answer.

The UD-Q3 results (Finding 5) show that layer-wise quantization fixes this failure.

Table 3: Retrieval task score degradation for Qwen3, where Q4 is worse in every task. This pattern is absent in Qwen3.5.

Task @ Context	**Q8**	**Q4**	**Q3**
S-RT @ 4k	100.0%	94.4%	96.7%
S-RT @ 8k	98.9%	84.4%	93.3%
M-RT @ 16k	96.2%	76.9%	81.8%

4.3 Finding 3: Short-Context Controls Confirm Domain Specificity

WikiText-103 perplexity and GSM8K accuracy (Tables 4 and 5) confirm that long context failures are specific to the retrieval tasks rather than all round capability degradation. Both models degrade smoothly in perplexity with decreasing precision: Qwen3.5 increases 0.80 pp (Q8→Q3), Qwen3 increases 2.11 pp. On GSM8K, Qwen3 outperforms Qwen3.5 substantially at full precision (93.6% vs. 85.6%), demonstrating that it is not entirely worse.

Table 4: WikiText-103 perplexity (lower is better). Both models degrade smoothly with no Q4 anomaly.

Model	**PPL Q8**	**PPL Q4**	**PPL Q3**	**PPL UD-Q3**
Qwen3-4B	17.15	17.66	19.26	18.81
Qwen3.5-4B	14.80	14.95	15.60	15.41

Table 5: GSM8K accuracy at short contexts. Qwen3 outperforms Qwen3.5, confirming it is not globally inferior.

Model	**GSM8K Q8**	**GSM8K Q4**	**GSM8K Q3**	**GSM8K UD-Q3**
Qwen3-4B	93.6%	92.0%	90.0%	88.4%
Qwen3.5-4B	85.6%	84.0%	87.6%	88.0%

4.4 Finding 4: Behavioral Divergence in Refusal Rates and Failure Modes (RQ3)

The two models undergo different behavioral failure modes under quantization pressure as context increases. At 16k, Qwen3 declines to answer half of all multi-fact questions. Qwen3.5 maintains a stable refusal rate regardless of quantization level or context length, with no escalation (Q4: 10.0% at 4k, 15.6% at 8k, 16.7% at 16k).

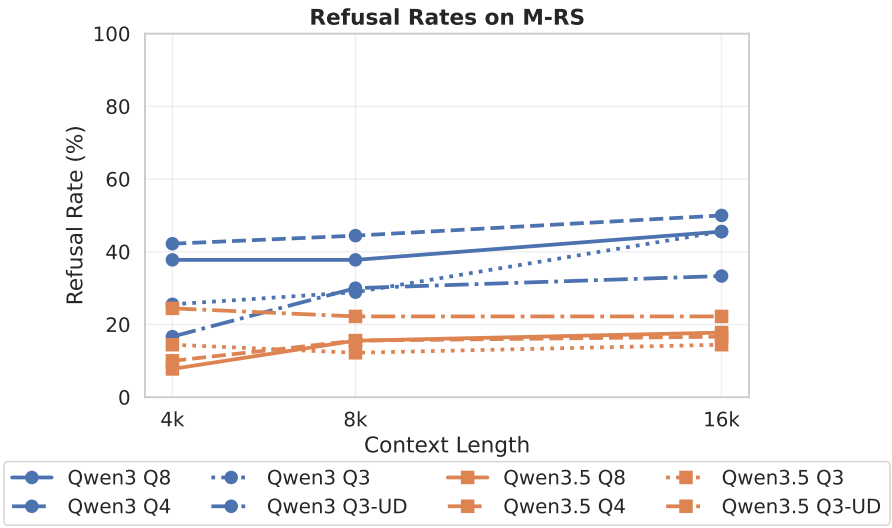

Figure 1: Qwen3.5 maintains a refusal rate of approximately 14%,while Qwen3 starts with a 40.4%, increasing to 50% as context grows.

Depth-position analysis at 16k M-RT reveals lost in the middle pattern for Qwen3 Q4: accuracy falls to 65% at depth 30% and 67% at depth 50%, recovering to 97% at depth 90% (Table 6; individual cells $n = 15$, ±5–7 pp). Qwen3.5 Q4 maintains 99% at every depth without exception, entirely absent the U-shaped degradation.

Table 6: M-RT accuracy by needle depth at 16k context. Qwen3 Q4 shows the "lost in the middle" trough at depth 30–50%. Qwen3.5 Q4 and Q8 are flat at 99% across all depths.

Depth	Qwen3 Q8	Qwen3 Q4	Qwen3 Q3
5%	95%	77%	81%
10%	100%	81%	77%
30%	95%	65%	73%
50%	93%	67%	72%
70%	95%	73%	87%
90%	100%	97%	100%

4.5 Finding 5: UD-Q3 Over-Performance as Evaluation Artifact and the Prompt-Ablation Result

Qwen3 UD-Q3 achieves 60.7% on M-RS overall, exceeding the Q8 baseline of 48.9% by 11.8 pp, a paradoxical result where a more aggressive compression outperforms high-fidelity weight. Aggressive quantization degrades instruction-following, causing the model to attempt answers instead of producing structured "NOT FOUND"

refusals. The partial-credit composite metric rewards guesses above the 0.5 similarity threshold, while a structured refusal receives zero credit. The UD-Q3 "improvement" is then an evaluation artifact, higher scores driven by increased guessing, not improved reasoning. UD-Q3 reduces Qwen3's M-RS refusal rate from 40.4% to 26.7% (Table 1).

Model	Quant	Total	Correct%	Refusal%
Qwen3	Q8	270	48.9	40.4
Qwen3	Q4	270	45.6	45.6
Qwen3	Q3	270	54.1	33.0
Qwen3	UD-Q3	270	60.7	26.7
Qwen3.5	Q8	270	77.4	13.7
Qwen3.5	Q4	270	74.8	14.1
Qwen3.5	Q3	270	73.7	13.7
Qwen3.5	UD-Q3	270	68.9	23.0

Table 7: M-RS Refusal Percentages by Model and Quantization Type

A prompt-ablation experiment provides the most novel and generalizable finding of this study. Removing the "NOT FOUND" instruction from Qwen3 Q8 M-RS prompts, forcing the model to attempt retrieval rather than abstain, caused it to successfully retrieve facts it had previously refused. This demonstrates that a portion of Qwen3's long-context failure reflects behavioral over-caution rather than genuine capacity exhaustion. This finding has direct implications for long-context evaluation design: benchmarks that include structured refusal instructions may interact with quantization and context length to produce measurement confounds, particularly when partial-credit metrics reward guessing over abstention.

5 Discussion

5.1 The Confounding Variables

The most important constraint on interpretation is that architecture and training vary. Qwen3.5 differs from Qwen3 not only in its GDN layers but also in training, RLHF fine-tuning procedures, and instruction templates. Its consistently lower perplexity (14.80 vs. 17.15 at Q8) indicates a stronger base model. Its substantially higher M-RS accuracy at Q8 (77.4% vs. 48.9%) suggests differences in reasoning-oriented training independent of architecture. A better-trained model may require less precision to maintain retrieval behavior simply because relevant weight patterns are more sharply

defined and less vulnerable to quantization effects.

This confound cannot be resolved within this study. The appropriate claim is that Qwen3.5 exhibits substantially greater quantization resilience than Qwen3 under the specific conditions evaluated. Stronger causal claim would require a controlled ablation: a single base model checkpoint with GDN and attention layers systematically substituted while all training factors are held constant.

5.2 Practical Implications for Practitioners

For practitioners deploying models on consumer hardware at long contexts, the Q4 anomaly carries a direct practical implication: dynamic quantization schemes may be more suitable than standard schemes for retrieval-focused workloads. Qwen3 UD-Q3 achieves 99.3% S-RT, exactly matching Q8, by protecting sensitive layers at higher precision. This recommendation applies specifically to retrieval tasks; the UD-Q3 edge on M-RT reasoning is partially effected by model not following instructions as stated earlier.

We argue that refusal rates should be tracked independently of accuracy scores. The prompt-ablation experiment shows that long-context benchmarks including structured refusal instructions may systematically underestimate the retrieval capacity of models with conservative alignment tuning.

6 Conclusion

This study presented an empirical comparison of quantization resilience between Qwen3-4B and Qwen3.5-4B across 6480 NeedleBench evaluations at 4k, 8k, and 16k token contexts. Five findings emerged. (1) At Q4 precision, the GDN hybrid exhibits zero Multi-Needle Retrieval degradation from 4k to 16k while Qwen3-4B loses 18.0 pp. (2) Q4_K_M produces a failure mode for Qwen3 in which Q4 underperforms Q3, while such failure modes are not present in Qwen3.5. (3) Short-context controls confirm these failures are domain-specific: perplexity and GSM8K degrade smoothly without anomalies, and Qwen3 surpasses Qwen3.5 on mathematical reasoning (93.6% vs. 85.6%, both Q8). (4) The two architectures show different behavioral failures: Qwen3's refusal rate escalates to 50% at 16k under Q4, and depth-position analysis reveals a classic lost in the middle trough, both absent in Qwen3.5. (5) An evaluation artifact is identified in which UD-Q3 spuriously inflates M-RS scores by suppressing

structured refusals; the prompt-ablation experiment shows the Q8 baseline retains retrieval capacity that is behaviorally suppressed by long-context overload.

Finding 5's prompt-ablation is the most generalizable contribution. It demonstrates that a fraction of what seems to be quantization-driven failure is in fact behavioral problems, with implications for how long-context benchmarks should be designed and interpreted. Specifically, benchmarks that include structured refusal instructions (e.g., "say NOT FOUND if the answer is absent") may mix capacity measurement with alignment tuning. Future frameworks should track refusal rates independently, and consider prompt-ablation as a diagnostic to distinguish behavioral from capacity failures. The Q4 anomaly additionally warns practitioners against assuming that more bits always means better retrieval performance: for deployments on consumer hardware, dynamic quantization schemes such as UD-Q3 may outperform Q4_K_M on retrieval tasks, despite Q4's wider adoption and lower bit-width. Qwen3.5-4B does not show this phenomenon.

6.1 Limitations

Experiments carried out have two major limitations. All evaluations rely on two models with different pretraining, to isolate the compounding error variable one must train 2 models with identical pretraining that differs just in architecture. The other limitation is context length, while it is enough to see retrieval degradation even with 16k context, state of the art models are pushing 1 million tokens. We had to experiment around to find a benchmark that models didn't score 100%, and multi-hop/reasoning benchmarks may not always reflect the practicioners' needs.

6.2 Future work

It is clear that the next step of experimentation must be controlled ablation substituting GDN and attention layers within a single model checkpoint.

References

Daniel Han, M. H., & team, U. (2023). *Unsloth*. https://github.com/unslothai/unsloth

Dao, T., & Gu, A. (2024). Transformers are SSMs: Generalized models and efficient algorithms through structured state space duality. *International Conference on Machine Learning (ICML)*.

Gu, A., & Dao, T. (2023). Mamba: Linear-time sequence modeling with selective state spaces. *arXiv preprint arXiv:2312.00752*.

Li, M., Zhang, S., Zhang, T., Duan, H., Liu, Y., & Chen, K. (2024). NeedleBench: Can LLMs do retrieval and reasoning in 1 million context window? *arXiv preprint arXiv:2407.11963*.

Liu, N. F., Lin, K., Hewitt, J., Paranjape, A., Bevilacqua, M., Petroni, F., & Liang, P. (2024). Lost in the middle: How language models use long contexts. *Transactions of the Association for Computational Linguistics*, *12*, 157–173.

Mekala, A., Atmakuru, A., Song, Y., Karpinska, M., & Iyyer, M. (2025). Does quantization affect models' performance on long-context tasks? In C. Christodoulopoulos, T. Chakraborty, C. Rose, & V. Peng (Eds.), *Proceedings of the 2025 conference on empirical methods in natural language processing* (pp. 9422–9470). Association for Computational Linguistics. https://doi.org/10.18653/v1/2025.emnlp-main.479

Michalak, F., & Abreu, S. (2025). Some attention is all you need for retrieval. https://arxiv.org/abs/2510.19861

NVIDIA. (2025). Nvidia nemotron 3: Efficient and open intelligence [White Paper]. https://arxiv.org/abs/2512.20856

Qwen Team. (2026, February). Qwen3.5: Towards native multimodal agents. https://qwen.ai/blog?id=qwen3.5

Yang, A., & the Qwen Team. (2025). Qwen3 technical report. *arXiv preprint arXiv:2505.09388*.

Yang, S., Kautz, J., & Hatamizadeh, A. (2025). Gated delta networks: Improving Mamba2 with delta rule [arXiv:2412.06464]. *International Conference on Learning Representations (ICLR)*.

Zandieh, A., Daliri, M., & Han, I. (2025). TurboQuant: Online vector quantization with near-optimal distortion rate [arXiv:2504.19874]. *International Conference on Learning Representations (ICLR)*.

Appendices

A Code Repository

The GitHub repository of this project is available at:
https://github.com/emrehannn/Qwen3-3.5-Quant-Comparisons

B AI Usage Declaration

This paper was researched, written, and executed entirely by the authors. AI tools were used in a strictly limited capacity to support the presentation and implementation of our own original work, specifically in the following ways:

- **Language rephrasing:** AI tools were used for linguistic refinement of the Abstract, Introduction, Background, and Discussion, and for script assistance. No content, analysis, or experimental findings were AI generated.
- **Code debugging:** AI coding tools were consulted to resolve implementation errors in the evaluation pipeline, including context length configuration issues, Flash Attention integration, and chunked prefill setup for the RTX 4060.
- **Benchmark scaffolding:** AI tools assisted in setting up the pipeline for the benchmark for the NeedleBench evaluation loop and scoring functions (Levenshtein similarity, coverage ratio, exact match). The scoring logic and experimental design were conceived and validated entirely by the authors.

All experimental results, tables, findings, interpretations, and conclusions presented in this paper are solely the work of the authors. No AI tool was used to generate, fabricate, or substitute any empirical result or analytical claim.